



Semester 2 Report

Preparing to study the surface of Phobos using IDEFIX WheelCam images

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Abstract

The Martian Moons eXploration (MMX) mission, led by JAXA and scheduled for launch in 2026, represents a pivotal opportunity to advance our understanding of Mars' natural satellites, Phobos and Deimos. This research supports MMX's scientific goals by investigating the feasibility of using WheelCam imagery from the IDEFIX rover to analyze Phobos' surface properties. Two primary objectives are addressed: (1) estimating wheel sinkage from imagery as a means of determining regolith mechanical properties, and (2) evaluating the potential for spectral analysis using panchromatic images under narrow-band LED illumination. For sinkage analysis, a wheel detection and ellipse fitting algorithm was developed and tested under varying conditions, showing promising initial results, particularly with JAXA's regolith simulant. Spectral studies demonstrated meaningful correlations between image brightness and reflectance, validating the approach under controlled conditions. A custom testbed simulating Phobos-like conditions enabled robust method validation. The research highlights key next steps, including validation with geological analogs, calibration refinement, and algorithm optimization, to ensure reliable scientific return from the IDEFIX rover's imaging systems.

Keywords: Phobos, MMX, regolith characterization, planetary science, image processing, wheel sinkage, spectral imaging

Signed Declaration of Authenticity

This assignment is entirely our own work. Quotations from literature are properly indicated with appropriate references in the text. All literature used in this piece of work is indicated in the bibliography placed at the end. We confirm that no sources have been used other than those stated.

We understand that plagiarism (copy without mentioning the reference) is a serious examination offence that may result in disciplinary action being taken.

Date: June 24, 2025

Signature

F. Mellab
A handwritten signature in black ink, appearing to read 'F. Mellab', with a large, stylized flourish underneath.

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Introduction

Over the past century, humanity's exploration of the Solar System has made extraordinary advances. Despite this progress, many regions of space remain unstudied and undiscovered due to the limitations of our technology. The MMX mission (Martian Moons eXploration) is the first of its kind, in which the close observation of the unexplored Martian moons will become possible. The moons of Mars were first discovered in 1877 by Asaph Hall [1], named Phobos and Deimos, and their origin remains a debate up to this day, with three hypotheses that are all feasible under the current available data. The origin of the moons is believed to be either resulting from a giant impact, the capture of asteroids or co-formation [2],[3], further information on section 1.1.

Other missions have left Earth to explore the moons of Mars and failed. Now, set to depart in 2026, JAXA's (Japan Aerospace Agency) MMX mission aims to collect invaluable images and data from Phobos and Deimos, as well as collect samples that will be taken back to Earth for their study. Touch-and-go procedures, used in the past for sample-return missions, are not feasible due to engineering constraints [4]. Therefore, the MMX spacecraft will land on the surface of Phobos, the larger of the two moons, to take the samples. The surface conditions of Phobos are unknown and this poses a risk for landing. Consequently, it will first drop the IDEFIX rover from altitude to assess the risks on the ground. Through the IDEFIX rover and its different instruments, the surface properties of Phobos will be studied.

The focus of this project lies on whether these properties can be studied through the Wheel-Cams, panchromatic cameras that point to each of the wheels of the rover. More specifically, this report will study the feasibility of gathering relevant information through the observation of the wheel sinkage into the surface of Phobos and the use of different colored LEDs to obtain a spectral study of the surface. Both of these approaches remain in the initial stages and in this report, the advancements will be discussed, as well as the possible limitations of the procedures and conclusions drawn from the testing made throughout the project. Finally, the future of this investigation will be discussed, including the next steps and objectives.

1 Semester Section

In this chapter, the scientific background of our project will be presented. Section 1.1 will deal with the general context of the mission and how it relates to the goals that will be proposed for this research project. Section 1.2 will focus on the theoretical background, explaining the conceptual basis needed for the completion of the project as well as how this research will contribute to the state of the art in this domain. Finally, on section 1.3, the general objectives of the research will be discussed as well as the objectives specific to this semester.

1.1 Context and key issues

1.1.1 Mission context

This research project focuses on the Martian Moons eXploration (MMX) mission, led by the Japan Aerospace Exploration Agency (JAXA) in collaboration with Centre National d'Etudes Spatiales (CNES) and the Deutsches Zentrum für Luft- und Raumfahrt (DLR). MMX will be the first sample-return mission from Phobos, the larger of Mars' two moons, and it will also conduct the most detailed investigation to date of both Phobos and Deimos.

Although past missions such as NASA's Viking program have gathered detailed images of the Martian moons [5], and others like Phobos 1, Phobos 2, and Phobos-Grunt have attempted direct investigation [6], only Phobos 2 succeeded in transmitting scientific data. These limited and incomplete missions have left many unanswered questions about the origin, composition and evolution of Mars' moons.

This knowledge gap has led to the proposal of several mission concepts over the years, but MMX remains the only currently approved mission to investigate the Martian moons [4]. Initially proposed in 2015, MMX is scheduled to launch in 2026.

After launch, the spacecraft will head towards Mars where, upon arrival, it will enter orbit around the planet. Afterward, it will transition into orbit around Phobos, where it will eventually land to collect surface material. Once the samples have been collected and the spacecraft commences its return journey, MMX will perform a close flyby of Deimos, to obtain high resolution images and gather spectral data.

Although Phobos remains unfamiliar in many aspects, some of its general characteristics have been established through terrestrial observations and previous Mars missions. These include estimates on its size, mass, orbit, surface gravity, and overall shape [7]. As a result, the design of the MMX spacecraft can begin with a set of relatively well constrained physical parameters of its target, unlike previous asteroid missions.

The main objectives of the MMX mission are focused on revealing the origins of Mars' moons by the study of their composition through both close-range observations and the return of physical samples. Due to the limited knowledge about the physical and mechanical properties of Phobos' surface, the safe landing to collect a sample poses significant challenges. To reduce this risk,

the mission includes IDEFIX, a small French-German rover designed to land on Phobos before the main spacecraft [8]. IDEFIX will conduct in-situ measurements to evaluate the surface conditions, determining whether it is safe for landing to occur and for the sample to be taken. Moreover, it will gather valuable information on the moon's surface characteristics, which will contribute to the overall scientific goals of the mission.

The rover is expected to be released from an altitude of less than 100 meters (ideally around 45 meters), descending freely without any braking mechanism until it impacts the surface, where it will right itself by repositioning its legs to achieve a stable orientation. The overall fall velocity will be low, as the gravitational acceleration on Phobos is about 0.0057 m/s^2 . Once upright, the rover will deploy its solar panels and begin its surface operations. However, the movement speed and the total area covered is unknown as it is constrained to the terrain conditions [8]. Although the mission aims for the rover to operate for approximately 100 days, actual performance will depend on surface conditions [9].

1.1.2 Mars' moons: Phobos

There are several competing theories regarding the origin of the Martian moons, however their exact formation remains unknown. Due to their shape and size, it is believed that they could be captured asteroids. Another theory suggests that they are the result of debris that had its origin on a giant collision between Mars and a protoplanet [10]. A third one has proposed that they are a result of co-formation, formed at the same time as Mars, however, the low density of the moons makes this theory the least probable of the three [11].

Phobos is the larger and innermost of Mars' two moons, orbiting at a gradually diminishing distance of approximately 6000 kilometers from the Martian surface. Due to the gravitational pull of Mars, Phobos' orbit is slowly decaying. Therefore, the moon is expected to either crash into the planet or break up into a ring in approximately 50 million years [12].

As mentioned in subsection 1.1.1, the general characteristics of Phobos are established with a certain degree of confidence. Its overall shape is very irregular due to the presence of several craters, and its mass is insufficient to make it a spherical body.

Regarding the composition of the surface, constraints exist, although with a high degree of ambiguity. The surface of Phobos has been studied through spectral imaging, and related to others of known composition and similar spectral characteristics. One of the most distinct characteristics of the surface of Phobos is the fact that it exhibits two clear geological units, red and blue, of which the origin is unknown, very close to its main crater Stickney. The spectral characteristics of these geological units have been studied and have proven to exhibit similarities; its origin is also a mystery that the MMX mission will address.

In general, Phobos presents a very dark spectra and very low albedo, comparable to P-type and D-type asteroids [13] but its spectral characteristics are compatible still with the both of principal formation theories.

The surface of the moon is known to be covered by regolith, which is most likely formed by the accumulation of impact debris on its surface [14]. The properties of the regolith evolve over the history of the body and can differ depending on the mineralogy present on Phobos,

impacts, thermal alterations and fatigue and space weathering. Space weathering is a process in which a body will darken or redden due to the pass of time, deepening the absorption bands in the optical spectra. It is the explanation for the differences that occur on bodies of seemingly similar composition but with different spectral properties [15].

Other properties, such as mechanical properties of the regolith are related to the granular behavior of the regolith and particle size. For these properties, theoretical estimations have been made due to limited knowledge on how granular materials behave on low gravity environments. The range of particle size is believed to be between 30 μm to 10 cm [7], however, it should be noted that given the concerns about the landing site, the images taken by IDEFIX will likely be biased towards finer-grained surface materials [16].

1.1.3 IDEFIX Rover: WheelCams

The IDEFIX rover has four main scientific instruments to obtain measurements from the surface of Phobos. These include two navigational cameras (NavCams), two cameras pointing towards the wheels (WheelCams), a Raman spectrometer known as RAX and a miniature radiometer called miniRad [8]. Each instrument serves a purpose and together they provide complementary data that will contribute to the understanding of the properties of the moon's surface.

This research project focuses mainly on the WheelCams. These cameras will observe the interaction between the wheels and the regolith, obtaining crucial information that will help determine the properties of the regolith. Their field of view is represented in red on Figure 1.1.

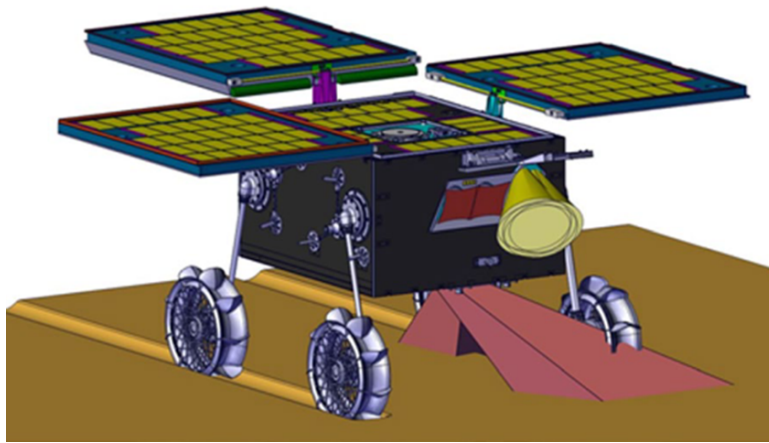


Figure 1.1: IDEFIX rover with deployed wheels and solar panels [8]

The WheelCams are based on a multi-purpose camera for Space exploration named CASPEX, which was developed by CNES since 2015[16]. The WheelCams have a bandwidth of 550 to 900nm, a field of view of 32.5° and take images of size 2048x2048 pixels. Each WheelCam is also be equipped with 7 LEDs, 4 White LEDs used for navigating and 3 colored LEDs (590 nm, 720 nm and 880 nm) to illuminate the environment.

The WheelCams are panchromatic cameras, in other words, they will receive information inside of a specific spectral range (corresponding to their bandwidth) and they will join said signal into

a single output, a grayscale image. The combination of these images with the LEDs will allow to obtain spectral information that will assist into the compliance of the mission objectives.

The specific objectives from the IDEFIX WheelCams are the following [16]:

1. Determine the physical properties of the regolith particles
2. Determine the bulk mechanical properties of the regolith
3. Determine the dynamical behavior of the regolith
4. Observe possible layering in the shallow sub-surface
5. Constrain the mineralogical composition of the surface material
6. Assess space weathering
7. Determine regolith geological classes
8. Constrain the absolute local gravitational acceleration

Once again, the completion of these objectives will be possible by studying the interaction between the wheel and the regolith and by evaluating the information obtained by the use of the LEDs. In order to comprehend the possible conditions of the regolith in Phobos as well as to study the feasibility of performing certain tests, a testbed was constructed to recreate the environment in Phobos.

1.1.4 Experimental testbed

The main objective of the testbed is to develop the image processing tools to prepare for the exploration of Phobos by the IDEFIX rover[16]. The testbed consists of three main sub-assemblies (the main structure, the material container, and the wheel assembly), multiple sensors (a motor encoder, two displacement sensors, and a laser profilometer), and two cameras. The two cameras are positioned to observe the wheel-regolith interactions from the same perspective as the front and rear IDEFIX WheelCams respectively. The cameras featured in the testbed are Qualification Models (QM) of the WheelCams installed on IDEFIX.

The testbed is an important tool in this research project to acquire images with which the image processing algorithms will be developed and to test the algorithm in a range of conditions expected on the Martian moon. In particular, the regolith simulant in the testbed can be changed to simulate different conditions. The vertical laser can be used to obtain the actual wheel sinkage value, which will be used to validate the results obtained from the image processing algorithm discussing in chapter 2.

The testbed, just as the IDEFIX rover, is also equipped with several LEDs for each camera. 2 white LEDs that will be used when the rover is in motion and 3 colored LEDs (with wavelengths of 590nm, 720nm and 880nm respectively) that will be used when the rover is static. Through the use of these LEDs and the panchromatic cameras, spectral studies will take place. The

testbed will be used in chapter 3 to perform experiments to understand if and how spectral information about the regolith can be extracted from the WheelCam images by comparing to results from a spectrometer.

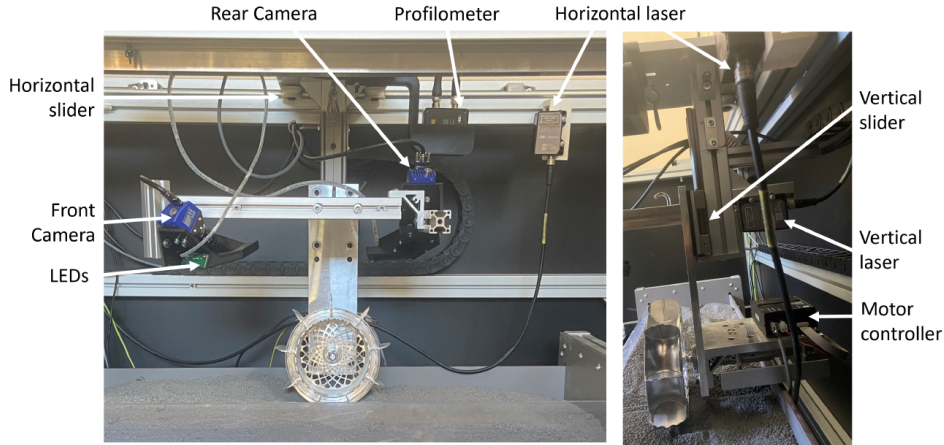


Figure 1.2: ISAE-SUPAERO WheelCam testbed [16]

1.2 State of the Art and novelty of this research

The state of the art in performing science with the WheelCam images is advancing on many fronts. Image processing algorithms are currently being developed at ISAE-SUPAERO to support characterization activities in preparation for receiving the first images from the IDEFIX rover on Phobos, aiming to provide insights into each of the WheelCam science objectives presented in subsection 1.1.3. Some examples of the analyses being developed include studies on the trench morphology left by the rover's tracks, which hold information about regolith cohesion. Work is also being done to study the particle size frequency distribution (SFD) of the regolith, which will reveal information about processes such as impacting and transporting experienced by the surface material[16].

In terms of studying sinkage, students at ISAE-SUPAERO have already conducted research to develop an image processing algorithm to detect the wheel and estimate sinkage[17]. Their work focused on developing a Machine Learning (ML) algorithm capable of identifying the wheel contour. Their work also included developing an algorithm to perform a homography transform, which is needed to compensate perspective effects from the camera and retrieve the profile of the wheel as seen from a perpendicular direction. Finally, their work included writing an algorithm for fitting an ellipse to the wheel contour.

On the topic of performing spectral analysis using the WheelCam images, no prior research has been carried out specifically on this issue. However, there is evidence from literature of similar research being carried out. In particular, studies have been done on reconstructing hyperspectral images using LED illumination in narrow wavelengths from a monochromatic camera[18]. Their research demonstrates that spectral information can be extracted from a series of monochromatic images illuminated by RGB LEDs of different wavelengths[18].

1.3 Aims and Objectives

The goal of this research project is to prepare for the scientific analysis of WheelCam images from Phobos by developing automated, reproducible tools capable of extracting wheel sinkage and spectral information. These tools must be robust and feasible considering the time and operational constraints of the MMX mission. To ensure reliability, the methods will be tested across a broad range of simulated conditions that mimic the expected environment on Phobos.

The specific objectives for this research are:

- To design an algorithm that can accurately identify the wheel, detect the regolith–wheel interface, and quantify the depth of sinkage.
- To assess whether the material composition of the regolith can be determined from the response of the regolith to illumination by LEDs at specific wavelengths.

These objectives directly support the WheelCam science goals, as outlined in subsection 1.1.3:

- **Sinkage estimation:** contributes to science objectives 1 (determining physical properties of regolith particles) and 2 (determining bulk mechanical properties of regolith).
- **Spectral analysis:** contributes to science objectives 5 (constraining mineralogical composition) and 6 (assessing space weathering).

As this research project runs for one year, the objectives at this stage of the research are more restricted. The semester objectives are:

- To improve the wheel detection and ellipse fitting algorithms developed by previous students, ensuring robust and accurate identification of the wheel across a wide range of WheelCam images as a first step towards quantifying wheel sinkage.
- To develop a simplified model to study the possibility of extracting spectral information from panchromatic images illuminated using the three colored LEDs present on the IDEFIX rover.

2 Wheel Sinkage

2.1 Introduction

In this chapter, an automatic algorithm for estimating wheel sinkage is developed. Wheel sinkage, defined as the vertical displacement of a wheel as it penetrates the soil, can provide insight into various physical properties of planetary regolith. Indeed, wheel sinkage is related to the soil's bearing capacity (ie. the maximum pressure it can withstand before undergoing shear failure) and regolith mechanical characteristics including cohesion and friction angle[16]. These properties are important to quantify as they influence the rover's locomotion and the landing dynamics for the main MMX spacecraft, which will land on Phobos in the weeks following the beginning of IDEFIX's exploration. The information obtained from sinkage analysis, along with all other properties studied under the WheelCam science objectives, will contribute to determining the geological classification of the regolith[16].

2.2 Theoretical Background

Measurements of wheel sinkage and slip can yield valuable insights into the mechanical properties of the regolith. However, since the IDEFIX rover does not provide telemetry on motor torque or current, these parameters will be derived exclusively from WheelCam imagery [16].

A modeling framework must be established to link observed wheel behavior to the underlying physical and mechanical characteristics of Phobos' regolith. By fitting semi-empirical terramechanics models (such as the Bekker or Reece-Wong models) to the observed sinkage and slip data, it is possible to estimate key mechanical properties of the regolith, including cohesion, internal friction angle, shear strength, and stiffness [16].

However, these models must be adapted to the low-gravity environment on Phobos to provide accurate results. Low gravity reduces the normal force acting on the wheels and affects the behavior of the regolith as it interacts with the wheels. As a result, significantly higher wheel slippage is expected to occur in this type of environment and model parameters must be adjusted accordingly[16].

The bearing capacity of the regolith, a property crucial for predicting spacecraft landing dynamics, will be estimated independently using both direct sinkage measurements and analysis of the trench morphology left by the rover's tracks [16].

2.3 Methodology

In this section, the development process of the wheel detection and sinkage estimation algorithm will be presented. The discussion will begin with an explanation of the rationale behind the chosen image processing approach. Then, a general overview of the algorithm pipeline will be

provided to provide a system view. Finally, each component of the algorithm will be examined in depth, highlighting the specific methods and techniques used at each stage.

2.3.1 Selection of image processing approach

At the beginning of the project, a wide range of image processing techniques were explored to determine the most effective way to approach the problem of detecting the edge of the wheel captured by the WheelCams. These processing techniques can be classified in two distinct approaches: Machine Learning (ML) and classical Computer Vision (CV) based methods. These routes differ fundamentally in their underlying philosophy and in their scope of applications.

In general, ML models for image processing require a large database of labeled images to train the model to be sensitive to relevant features present across the images. This allows the algorithm to generalize well to unseen data. This approach is often preferred in the context of complex images, where classical tools are not able to identify and extract meaningful features.

Algorithms built on classical CV techniques can be developed with a limited set of data and offer the advantage of being interpretable, allowing for fine-tuning of specific parameters. However, these algorithms require careful design considerations to ensure robust performance in the full-range of expected conditions.

In the context of this project, it was decided that classical CV techniques will be utilized, mainly considering the limited set of images available, the difficulty of labeling images and the importance of algorithm interpretability to ensure robustness and the ability to effectively tweak the model considering the time constraints of the IDEFIX's mission on Phobos. As mentioned in section 1.2, a ML based approach was explored by previous students to perform image segmentation and identify the wheel contour. However, this method yielded unsatisfactory results due to the issues presented above.

2.3.2 Algorithm pipeline

The algorithm developed to detect the edge of the wheel is based on the concept of image segmentation to extract a binary mask of the wheel, followed by contour detection techniques to identify the edges of the mask and extract the contour of the wheel. Finally, an ellipse fitting algorithm is applied to remove noise from the contour of the wheel and close the wheel ellipse. The next step to be implemented will be the detection of the wheel/regolith interface, needed to estimate the wheel sinkage. The image processing functions used in this algorithm were taken from Python's "Open Computer Vision Library" (OpenCV) [19].

A thorough analysis of the algorithm's performance on unseen data will be important to assess the range of application of this algorithm, to ensure image processing operations run smoothly when the first images are acquired by IDEFIX on Phobos.

The image shown in Figure 2.1 taken using the testbed with the JAXA regolith simulant will be used throughout this chapter to illustrate the effect of the various processing steps in the pipeline.

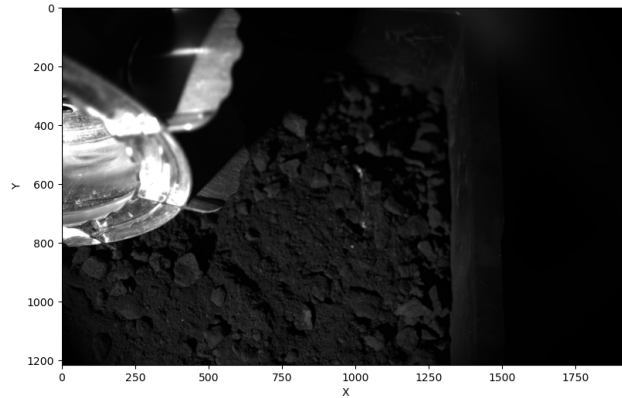


Figure 2.1: Reference testbed rear WheelCam image with JAXA regolith simulant

2.3.3 Image segmentation

In this subsection, the pipeline for the image segmentation part of the wheel detection algorithm will be broken down, highlighting the main processing steps and the key decisions that shaped the algorithm.

Data preparation

First, the image to be analyzed is read and stored as a NumPy array containing the value at each pixel in a range from 0 to 255. A check is performed to ensure the image is in grayscale format and has the expected dimension (1216x1936).

Segmentation threshold definition

The definition of a threshold for binary segmentation is a key step in developing the algorithm. For this application, global thresholding was used as opposed to local (or adaptive) thresholding because the illumination intensity is constant throughout the image, except for a bright object in the foreground.

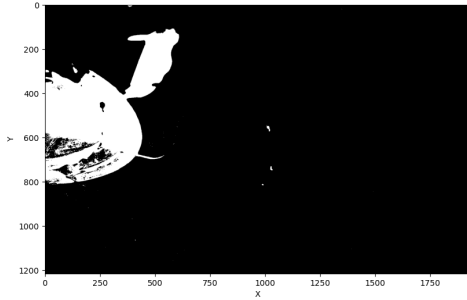
Different methods exist to define a global segmentation threshold. One of the most widely used is known as Otsu thresholding. This technique determines an optimal threshold such that the variance between the two classes (ie. interclass variance) of pixels is maximized[20]. The advantage of this method is that the bounds are determined using the image itself, no additional information must be provided. This property is desired because it keeps the processing pipeline deterministic and avoids the possibility of introducing human bias.

Another method to define the threshold for binary segmentation is by using adaptive bounds[21]. The threshold is defined based on the average pixel value within a certain "wheel confidence area": a small square area of the image where the wheel is expected to lie in all images[21]. A "wheel confidence area" of 200x200 pixels is defined as a location in the image that is expected to be fully within the wheel boundaries. The average value of a pixel is obtained within this

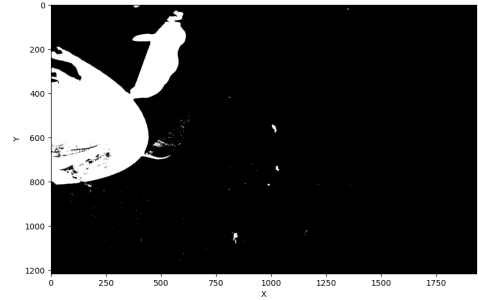
area to serve as a basis for thresholding. The thresholding boundaries are then defined as $[P_a \cdot \text{Average}, P_d \cdot \text{Average}]$, where P_a and P_d are the tuning parameters for the segmentation bounds. Pixels whose value lies inside this range are set to 255 and all other pixels are set to 0 to obtain a binary segmentation.

The parameters P_a and P_d were selected for the reference image by trial and error, the best performance was obtained by setting $P_a = 0.35$ and $P_d = 1.85$. This offered a good compromise between keeping most of the wheel in the mask while not including too many elements from the background.

Both global thresholding methods were implemented to compare their performance on a set of test images. An example of the segmented images obtained is shown in Figure 2.2. Visual inspection of these images lead to the conclusion that using adaptive bounds is preferred, as Otsu thresholding removed some parts of the wheel edge and caused more noise in the mask as seen in Figure 2.2a compared to Figure 2.2b.



(a) Binary segmentation using Otsu thresholding



(b) Binary segmentation using adaptive bounds with $P_a = 0.35$ and $P_d = 1.85$

Figure 2.2: Comparison of binary segmentation by Otsu thresholding and adaptive bounds

Morphological transformations

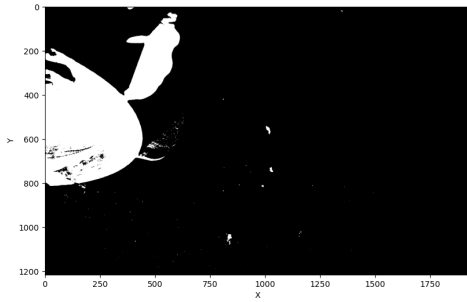
To improve the quality of the binary mask of the wheel and facilitate the fitting of an ellipse to the wheel edge at a later stage, morphological transformations should be applied. In particular, the protruding wheel grousers should be eliminated from the mask and the noise present in the foreground object should be removed as much as possible.

The two morphological transformations that were considered are OpenCV's opening and closing functions[22]. The opening function allows to remove small protrusions from a main shape, and works on the principle of erosion followed by dilation. The closing function is used to remove noise from the foreground object and works on the principle of dilation followed by erosion. For both of these operations, a key parameter to tune is the size and shape of the kernel.

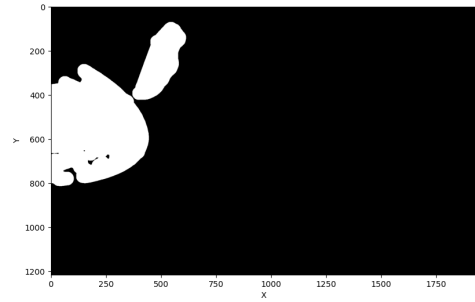
The choice of kernel allows to tune the transformations to act on objects of a certain dimension. Since the wheel grousers are large, an elliptical kernel of dimension (60, 60) was chosen for the opening function. The dimension of the noise elements being smaller, an elliptical kernel of

dimension (6,6) was selected for the closing function. Note that the shape of the kernel is an ellipse inscribed within a rectangle of the specified size, so in this case the kernel is circular.

Through trial and error, the optimal order to perform the morphological operations is to apply one iteration of the closing function followed by one iteration of the opening function. The effects of these transformations on the binary mask are shown in Figure 2.3. As can be seen, the grouser has successfully been separated from the wheel, the noise within the mask has been reduced and the noise in the background has been eliminated completely. However, this comes with a slight reduction in edge accuracy and the loss of some parts of the wheel.



(a) Binary segmented image before applying morphological transformations



(b) Segmented images after one iteration of closing and one iteration of opening

Figure 2.3: Effect of morphological opening and closing operations on the binary segmented image

2.3.4 Contour detection

After the binary mask has been cleaned, edge detection is performed to extract the boundary of the mask. First, Gaussian blur is applied to the masks with a 5x5 square kernel. This operation is important to ensure the stability of the subsequent edge detection, ensuring that edges are not omitted in the process. Then an edge map is obtained by performing Canny edge detection with sensitivity parameters $\text{minVal} = 220$ and $\text{maxVal} = 250$. OpenCV's Canny edge detector treats pixels whose value is above maxVal as "sure edges" and pixels below minVal as "sure non-edges" [23]. Pixels whose intensity lies between the two bounds are classified as edges only if one of the neighboring pixels is a "sure edge".

After generating the edge map, continuous contours are extracted using OpenCV's `findContours()` function. Each detected contour is then evaluated using the `minEnclosingCircle()` function, which computes the smallest circle capable of fully enclosing the contour [24]. Among these, the contour corresponding to the wheel is identified as the one whose enclosing circle has the largest radius. Initially, filtering based on the enclosed area was attempted, but this approach proved unstable and less reliable than radius-based filtering.

The points defining this contour along with the model parameters are finally stored as a NumPy array for further processing and analysis. The continuous contour obtained for the reference image after the filtering process is shown in Figure 2.4.

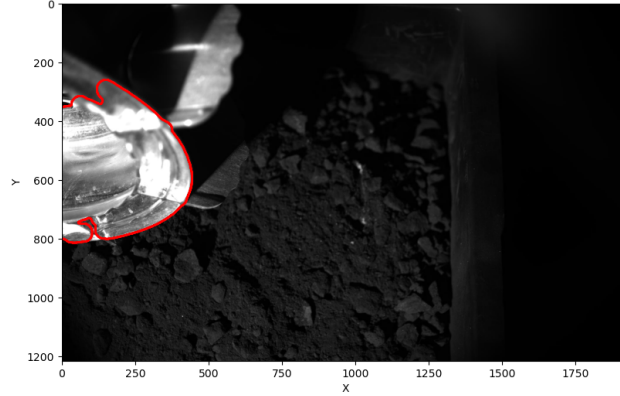


Figure 2.4: Contour selected after filtering continuous contours based on the maximum radius of their minimum enclosing circles

2.3.5 RANSAC based ellipse fitting

The shape defined by the circular wheel is an ellipse due to perspective effects from the camera position. The next step in the algorithm pipeline is thus to fit an ellipse to the detected contour. Although applying morphological transforms reduced the number of points that lie outside the wheel edge, there are still a number of outliers that must be ignored when fitting the ellipse. One way to deal with outliers in a robust way is to use a Random Sampling Consensus (RANSAC) algorithm.

A RANSAC algorithm is a type of iterative outlier detection algorithm. This algorithm determines an ellipse model that best fits a random subset of points in the contour and evaluates the quality of the fit by calculating the number of points that are within a certain distance to this ellipse model. This process is repeated for different samples of points, keeping the ellipse model with the most inliers until a limit on the number of iterations is reached or until the number of inliers for a given ellipse model is enough to assert that this ellipse model correctly fits the wheel contour.

The general equation for an ellipse is shown in Equation 2.1. This equation in five variables requires a minimum of five points to solve. If random samples of more than five points are used to fit a conic section model, the algorithm uses a MSE minimizer to find the best fit for that sample of points.

$$Ax^2 + By^2 + Cx + Dy + E = 0 \quad (2.1)$$

Understanding the behavior of the RANSAC algorithm is important for identifying scenarios where it may fail and how to avoid these. The required number of RANSAC iterations (k) can be estimated by analyzing the probability of selecting a set of inliers in a single iteration. Let w denote the proportion of inliers in the data, n the minimum number of points needed to fit a model (for an ellipse, $n=5$) and p the desired probability of selecting at least one set of only inliers after k iterations. The relationship between k and w is given by Equation 2.2 [25].

$$k = \frac{\log(1 - p)}{\log(1 - w^n)} \quad (2.2)$$

In Figure 2.5, this relationship is shown for $p=0.99$ and $n=5$. It can be seen that the number of iterations required grows asymptotically as w falls (ie. the noisiness of the contour increases). This highlights the importance, in particular, of removing the grouser from the binary mask as discussed in subsection 2.3.3 to obtain a clean contour.

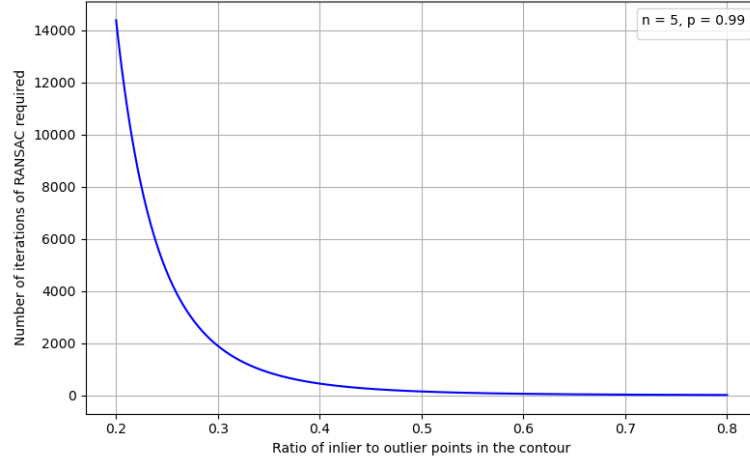


Figure 2.5: Number of RANSAC iterations required (k) as a function of the ratio of inliers to outliers (w) for $p=0.99$ and $n=5$

A RANSAC algorithm developed by previous students was applied to the contour, using random samples of five points [17]. The ellipse model shown in Figure 2.6 was found with 994 inliers out of a total of 1324 points in the contour. On closer inspection, the fitted ellipse closely follows the wheel edge on the right side but slightly cuts into the wheel at the top and bottom, suggesting the ellipse is slightly too flat in those areas

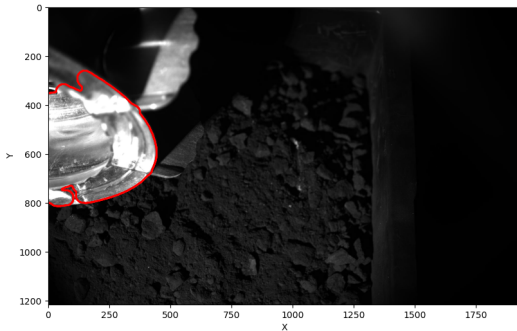


Figure 2.6: Ellipse fit to the wheel contour using RANSAC algorithm

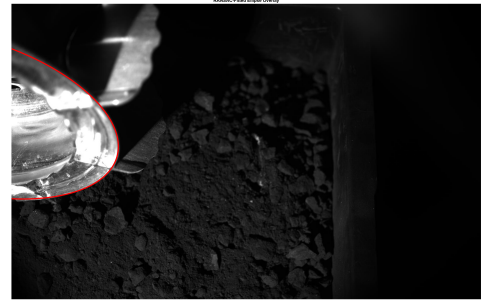
2.4 Results and Discussion

The wheel detection and ellipse fitting algorithms detailed in section 2.3 must be tested thoroughly to assess the range of conditions for which the method is valid and works as expected. This is especially important considering the degree of uncertainty regarding the Phobos surface, including the reflectivity of the regolith, its size and shape. In this section, the algorithm will be applied to a set of images taken with the ISAE testbed presented in subsection 1.1.4 using different regolith simulants and different levels of wheel sinkage. All the images presented here were taken with the testbed in the dark, with the scene illuminated by white LEDs to replicate the driving conditions on Phobos.

2.4.1 Rear WheelCam



(a) Wheel contour detected under JAXA regolith simulant conditions



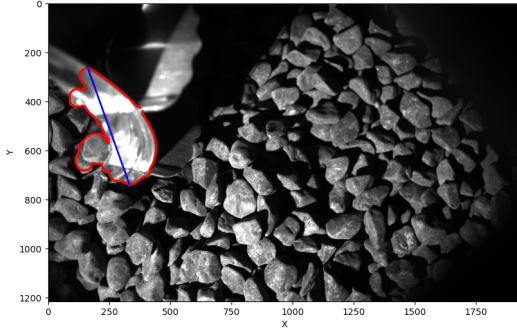
(b) RANSAC ellipse fit for the contour detected in Figure 2.7a

Figure 2.7: Results for the testbed rear WheelCam using JAXA regolith simulant

The image presented in Figure 2.7 is the base case for which the algorithm was developed and tuned. Thanks to the high contrast between the regolith and wheel, a large portion of wheel edge is detected by the algorithm.

It is important to note that this case corresponds to the most likely scenario of the conditions IDEFIX is expected to encounter on Phobos. Despite the uncertainty regarding the regolith on Phobos, analysis of remote images taken during flybys by previous missions (including ESA's Mars Express mission in 2004) has led scientists to believe that the regolith is likely dark and fine-grained, as modeled by the JAXA regolith simulant[16].

The image presented in Figure 2.8 presents a more challenging environment for the wheel detection process. The contrast between the wheel and gravel is less distinct due to the reflective nature of the gravel. There is also significantly more sinkage, making less of the wheel visible in the image. To obtain this result, some parameters were adjusted to adapt the model to the different conditions compared to the JAXA regolith images. In particular, the location of the wheel confidence area was adjusted (a square of 200x200 pixels with the origin at (200,400) was chosen).



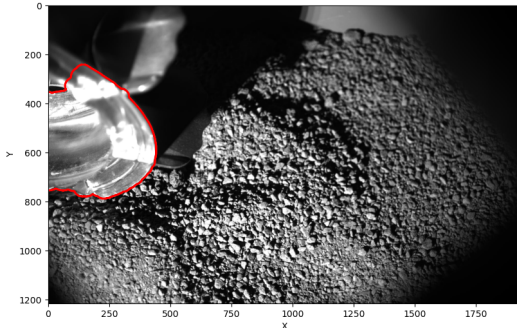
(a) Wheel contour detected under coarse gravel regolith conditions



(b) RANSAC ellipse fit for the contour detected in Figure 2.8a

Figure 2.8: Results for the testbed rear WheelCam using coarse gravel regolith simulant with significant sinkage

Moreover, the detected contour contains a high ratio of outliers to inlier points, which presented challenges for the RANSAC algorithm. To ensure a stable ellipse fit, additional filtering of the contour was necessary. Specifically, all points to the left of the blue line shown in Figure 2.8a, which passes through the detected extremities of the wheel edge, were removed prior to applying RANSAC.



(a) Wheel contour detected under Moon regolith simulant conditions



(b) RANSAC ellipse fit for the contour detected in Figure 2.9a

Figure 2.9: Results for the testbed rear WheelCam using Moon regolith simulant

The image presented in Figure 2.9 also poses significant challenges for the detection algorithm. Compared to the JAXA simulant, the moon regolith is much more reflective. To obtain this result, the same wheel confidence area as for Figure 2.8 was chosen. Additionally, the kernel for the morphological opening transformation was adjusted to a slightly larger (65x65 elliptical) kernel in order to filter out the grouzers as desired. Note that no other model parameters were altered (including the segmentation bound parameters P_a and P_d).

These three cases show that the model is able to detect the wheel contour with different regolith types that IDEFIX may encounter on Phobos. It is promising that the segmentation still

identifies the wheel as the main object when encountering different levels of reflectivity of the regolith. Only small modifications of the model parameters, in particular the wheel confidence area and kernel size were made accross the three cases which shows that the model is general and can be adjusted easily, which is important to ensure a smooth process when the first images arrive from Phobos. This highlights a key advantage of following a classical computer vision approach as opposed to a machine learning approach: results can be visualized at any step of the process to target the parameters that require tuning. A machine learning algorithm working as a black box is risky as it might fail under some conditions and tuning the model at that point would be very difficult, if not impossible.

The quality of the contours detected was sufficient for the JAXA simulant and Moon regolith cases to fit an ellipse accurately to the wheel edge. Regarding the gravel regolith with high sinkage, the quality of the edge was not sufficient to ensure a stable fit and further processing was required to obtain a good ellipse fit.

Note, however, that for now the quality of the ellipse fit is only assessed visually, but in the future when sinkage calculations are implemented in the pipeline, data from the lasers on the testbed will be used to objectively evaluate the performance of the algorithm. At that point, it will be interesting to explore the possibility of deriving a requirement on the quality of contour detection or ellipse fit based on the desired accuracy level of the final wheel sinkage measurement.

2.4.2 Front WheelCam

To evaluate whether the segmentation algorithm is applicable to both camera views, it must also be tested on images captured by the testbed front WheelCam. Figure 2.10 shows the detected wheel contour for a reference image from the testbed front WheelCam.

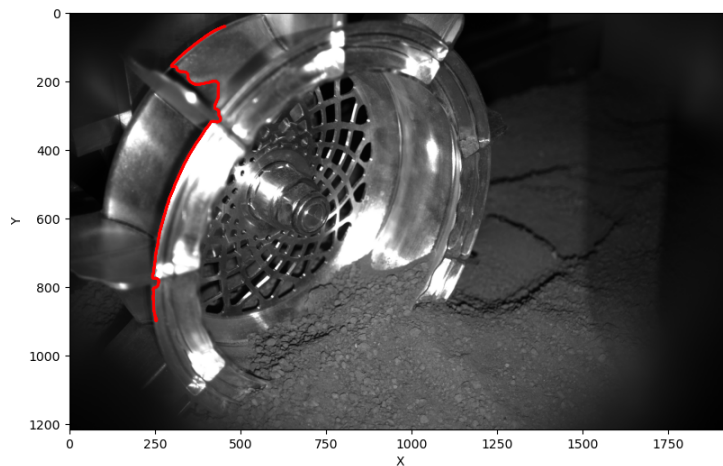


Figure 2.10: Wheel contour detected for the reference image of the testbed front WheelCam

As illustrated in Figure 2.10, the algorithm successfully identifies part of the wheel contour. However, when this contour is passed to the RANSAC-based ellipse fitting algorithm, it fails

to produce a stable result. This is likely because only one side of the wheel edge is detected, resulting in an insufficient distribution of contour points for accurate ellipse fitting.

In future work, the algorithm should be adapted or optimized specifically for the front WheelCam. It is possible that a single segmentation strategy may not perform well to both camera perspectives. If so, separate algorithms tailored to the front and rear WheelCam configurations should be developed.

2.5 Future Work

The next phase of this research will focus on integrating sinkage estimation into the image processing pipeline, following the methodology developed by previous students [17]. In their approach, wheel sinkage is estimated by applying a homography transformation to the ellipse detected via RANSAC, effectively restoring the original circular shape of the wheel as seen from a perpendicular viewpoint. The vertical distance between the center of this reconstructed circle and the regolith surface is then computed. By comparing this value with the known radius of the wheel, the sinkage can be quantified as a fraction of the radius.

Another important step will be to formally validate the rear WheelCam detection and fitting algorithms developed this semester. This validation process will involve testing the algorithm under varying signal-to-noise ratio (SNR) conditions, achieved by adjusting the LED power levels, as well as evaluating its performance across a broader range of regolith types and sinkage levels.

For front WheelCam images, further optimization is needed to ensure robust contour detection and reliable ellipse fitting under variable conditions. One potential method to explore is to divide the image in sections, apply different segmentation bounds for each part of the image and patch the obtained contours.

Finally, it is essential to define clear performance requirements for the complete sinkage estimation algorithm. This includes specifying the desired accuracy of sinkage estimation and exploring whether the corresponding accuracy thresholds can be defined for contour detection and ellipse fitting to meet that target. Establishing these metrics will help guide future improvements and validation of the overall system.

3 Material Composition Detection

3.1 Introduction

This experiment aims to validate whether differences in material properties can be detected using the IDEFIX rover's panchromatic camera under controlled LED illumination. This method relies on the use of the IDEFIX rover's panchromatic WheelCams and three LEDs of wavelengths 590 nm, 720 nm, 880 nm, to illuminate the surface and measure the relative brightness of the reflected light.

Given that the surface material on Phobos is expected to be composed of multiple elements and varying in grain sizes as explained in subsection 1.1.2, the spectral response will be affected not only by composition but also by surface texture. Therefore, to manage the complexity and test the feasibility of this approach, an initial simplified experiment is conducted using colored paper as a starting substitute for different materials. By comparing brightness values obtained from the images with known spectral curves for colored paper samples, we aim to test the feasibility of color differentiation based on brightness trends.

This simplified model helps to demonstrate whether differences in material properties can be detected through the brightness captured by the camera under specific LED illumination, and whether these results can be correlated with known spectral trends. It serves as a foundation toward more complex models that will ultimately aim for material identification on the surface of Phobos.

3.1.1 Objectives

Although the semester objectives have already been generally defined in section 1.3, let us specify exactly what the aims are:

- Perform a forward approach: write a model with which the use of the LEDs for spectral measurements can be validated
- Perform the experimental tests and evaluate the results. Critically state the assumptions made and limitations observed to the approach
- Perform a backwards approach: test whether the LEDs can be used for what the model proposed
- Evaluate its feasibility and state the modifications needed for the success of future testing

All of these objectives have been completed and will be discussed in detail on section 3.4 of the project.

3.2 Theoretical Background

3.2.1 Key Concepts

All of the definitions in the following paragraphs will be the definitions of the concepts in the specific application of this research project. Some concepts may have different definitions whether referring to radiometry, photometry, astronomy, among others.

This project focuses mainly on brightness and light intensity. Brightness will refer to the amount of "whiteness" on our grayscale images. It will be a value from 0 to 255, 0 indicating complete blackness and 255 indicating complete whiteness. This value will be measured directly from our image during the data processing. Light intensity, on the other hand, is obtained from a spectrometer and will be our comparison measurement for the brightness values. Light intensity refers to the signal strength or power received by the spectrometer's detector when observing a specific location. It is measured as the amount of light energy detected at each specific wavelength, providing information on how much light is absorbed, transmitted, or emitted by a sample at each point in the spectrum.

In addition to this project's focus on brightness and light intensity, it is important to consider reflectance and absorption. Reflectance is a proportion of light that a surface reflects relative to the total light received. It is broadly used in spectral studies and it is key in the interpretation of the brightness values that will be recorded on this project. Although it will not be computed per se, it will be referred to as a measurement related to brightness and fundamental for this investigation. Absorption refers to the proportion of light taken in by the material, rather than reflected. By identifying patterns in reflectance and absorption, material properties and composition can be identified and associated with said material. During this semester, the feasibility of identifying these properties through brightness is studied.

Another key concept will be sensitivity. Panchromatic cameras, as explained in subsection 1.1.3, receive information from various wavelengths, however this information is not always received equally. The sensor has a spectral sensitivity curve as illustrated in Figure 3.1, meaning it responds more to certain parts of the spectrum and less to others. This affects how the final image appears since brighter or darker areas may not only depend on how much light is present, but also on which wavelengths dominate the scene.

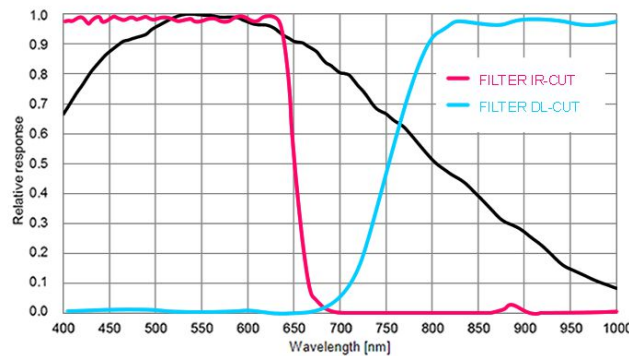


Figure 3.1: WheelCam's detector sensitivity in terms of wavelength [26]

3.2.2 Integration of key concepts

Recapitulating the theory explained in section 1.1, as the WheelCams are panchromatic, as well as the testbed WheelCams, there is no way to distinguish in which wavelength the data is received.

Combining the environment by illuminating it with a LED in a narrow spectral range will mean that the image received will consist of the response of the environment to said wavelength. In other words, a visual representation of reflectance and absorption at a concrete wavelength will be obtained.

Although this image is a visual representation of the spectral properties, the actual values of reflectance and absorption cannot be obtained through this process. Only brightness will be obtained from the grayscale image. However, brightness is proportionally related to reflectance. Therefore, conclusions about reflectance can be drawn from the values of brightness.

$$f(x, y) = i(x, y)r(x, y) \quad (3.1)$$

Equation 3.1 is obtained from [27], where it is explained that the brightness of an image $f(x, y)$ is related proportionally to reflectance $r(x, y)$ by illumination components $i(x, y)$. The latter is determined by the illumination source, whilst the reflectance is a property of the material under illumination. Moreover, as seen in subsection 3.2.1, the sensitivity of the camera's detector must be considered. Therefore, Equation 3.1 becomes Equation 3.2.

$$B(\lambda) = L(\lambda) \times R(\lambda) \times S(\lambda) \quad (3.2)$$

In this equation, the relationship between brightness and reflectance can be observed in terms of wavelength (λ), where $B(\lambda)$ depicts brightness; $R(\lambda)$, the reflectance; $L(\lambda)$, the illumination source intensity and $S(\lambda)$, the camera's sensitivity.

The illumination source intensity refers to how the light varies with wavelength. In this case, the LEDs used for the experiment function over a very narrow range of wavelengths with slight variation, therefore, the assumption of a constant value has been made. This assumption can lead to the increase in the error of the measurements, however it is an initial assumption that can be made for the simplified model and can be revisited later on.

The sensitivity is a factor that depends on the camera as presented in subsection 3.2.1, and has a value between 0 and 1. Since it varies with wavelength, it can be considered constant for the same LED but cannot be neglected for the comparison of the three LEDs. Therefore, it will only be relevant when comparing values from different LEDs.

As an approximation of Figure 3.1, the values of sensitivity factors for the LED's wavelengths are given in Table 3.1.

These are expected to be included in the camera's calibration, and already added to the grayscale image. The sensitivity for infrared is significantly low, which will influence our results later on.

Through the previous discussion it has been shown that, in theory, brightness holds a pro-

Wavelength	Sensitivity
590 nm	0.98
720 nm	0.75
880 nm	0.3

Table 3.1: Panchromatic’s camera sensitivity for a given wavelength

portional relationship to reflectance under controlled conditions. This allows brightness values obtained from the images to be used as indirect indicators of material reflectance. In order to draw significant conclusions, these values must be compared to experimental spectral studies of the desired material for testing.

For each available LED, an average brightness value will be obtained at small region of pixels. By analyzing the trend of these values across wavelengths and comparing them with known reflectance curves from spectrometric studies, the recognition or differentiation of different materials on the image is possible. In the case of this experiment, the comparative data will be obtained by the use of a spectrometer.

While the exact composition of Phobos’ surface materials is unknown, variations in brightness at specific wavelengths can still provide relevant information regarding space weathering and material composition as indicated in section 1.3. More specifically, in the wavelength range at which the cameras and LEDs operate, the images obtained can be use to reveal the presence of minerals such as olivine and pyroxene that can be detected in the 800 to 1000 nm region [16].

3.3 Methodology

Due to the complexity of the problem, it was necessary to begin testing with a simplified model to validate whether material differentiation is feasible using spectral methods. The initial idea was to use a solid sample of known spectral properties that would allow the direct comparison of the image data with the known reflectance data. However, the colored papers are an even more simplified approach that allows us to increase the number of test cases and therefore, the data obtained.

This approach allows for a straightforward experiment in which the brightness values from images of each colored paper, illuminated by each LED, are recorded. A spectrometer is used in parallel to provide a reference curve of light intensity versus wavelength for each sample and LED combination. Note that the spectrometer output is not a direct measure of reflectance, however, intensity values are also proportional to reflectance, allowing the comparison of spectral trends between both measurement systems. The trends of the data from the brightness in the images and the spectrometer values should share a significant resemblance for each colored paper used.

3.3.1 Experimental Test Set-up

The following equipment has been used to perform the experimental tests.

- ISAE-SUPAERO testbed (as described in subsection 1.1.4)
- Four monochromatic LEDs: Only one of the White (used as reference), Yellow (590 nm), Red (720 nm) and IR (880 nm).
- Panchromatic testbed WheelCams
- 11 colored papers in: Blue, Dark Blue, Dark Green, Dark Pink, Green, Orange, Pink, Purple, Red, White and Yellow.
- Spectrometer JETI LiVal V6.14.4

The setup consisted of placing each colored paper sample in the test area, on the location that simulates the regolith surface. The spectrometer was positioned over this area and fixed in a point illuminated by all three monochromatic LEDs. The spectrometer includes an integrated laser, which was used to visually identify the exact measurement point on the paper. This reference point ensured that all measurements were taken under illumination from each LED and allowed for a consistent measurement during image processing later on, even if the point chosen was not necessarily the location of maximum LED intensity on the paper. While this may slightly affect absolute intensity values, it reduces variability and improves comparability between measurements. This set-up can be observed on Figure 3.2.

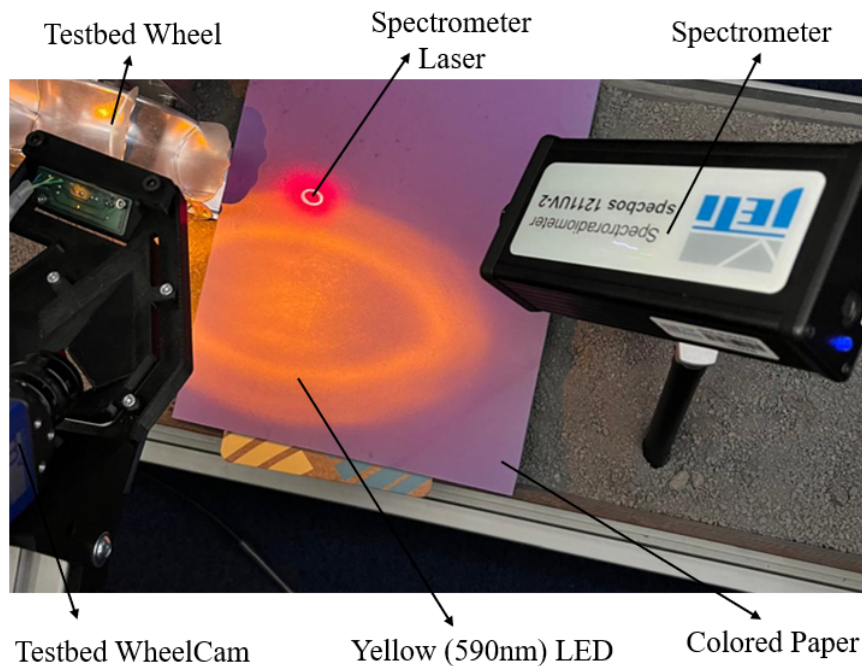


Figure 3.2: Experimental Set-Up example, with yellow LED on and purple colored paper

Once the spectrometer and sample were aligned, the testbed was sealed to eliminate ambient light interference. However, due to minor gaps in the enclosure, some ambient light contamination may have occurred. This must be acknowledged as a potential measurement error and a limitation of the experimental setup.

For each paper sample, four measurements were taken: one under each LED (White, 590 nm, 720 nm, and 880 nm), using both the panchromatic camera and the spectrometer.

3.3.2 Data Processing and Analysis

For each LED, a brightness value was extracted from the grayscale image using the average intensity of a 5×5 pixel area centered approximately on the point measured by the spectrometer. This point was identifiable through the spectrometer's built-in laser, which provided a visual reference. Using an area average helped reduce the impact of random noise and ensured consistency across all measurements.

The spectrometer provided a full spectral curve for each LED illumination, from which the value corresponding to the LED's wavelength was extracted. Although each LED was labeled with a nominal wavelength, slight variations in their emission spectra caused, in some cases, the peak light intensity to occur at a slightly different wavelength than the expected one. To account for this, the peak value was considered instead as it represented a more accurate measurement. This was evaluated experimentally.

This approach allowed for direct comparison between the trend of brightness values obtained from the images and the corresponding intensity values from the spectrometer, results found in Table 3.2. The objective was to evaluate the similarities between both trends and to validate the feasibility of using the combination of the LEDs and panchromatic camera to distinguish material composition.

In addition, the brightness values for each individual LED were compared in all paper samples, results found in Table 3.3. This provided additional insight into the consistency of the brightness trends and their accuracy, adding further comparisons to validate the correlation between the experimental brightness values and the known spectral characteristics of materials.

To analyze the trends, several approaches were used. To facilitate a visual comparison, both the brightness and spectrometer values were normalized and plotted together to observe the similarities between the curves.

Quantitative comparison was also performed using the Pearson Correlation coefficient and Cosine Similarity coefficient.

The Pearson correlation coefficient [28] measures the linear relationship between two sets of variables, however, it does not capture nonlinear or monotonic relationships. It is represented by the following formula:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (3.3)$$

where x_i and y_i are the individual values from the two datasets being compared (brightness and

intensity), and $\bar{x}$ and $\bar{y}$ are their respective means. The result r ranges from -1 to 1 , where:

$r = 1$ indicates perfect positive correlation

$r = -1$ indicates perfect negative correlation

$r = 0$ indicates no linear correlation

The Cosine similarity [29] measures the angle between two vectors, ignoring the magnitude. It ranges between 0 and 1, where the closeness to 1 indicates a more similar orientation. It is useful for comparing shape and pattern rather than scale however, it disregards magnitude which might influence our ability to distinguish between papers.

$$\text{CosSim} = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|} = \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2}} \quad (3.4)$$

where x_i and y_i are the individual values from the two datasets being compared (brightness and intensity).

The Pearson correlation coefficient is significantly useful in this case because it measures how two variables change in a linear and proportional manner. This is exactly what was expected of the results, to have a proportionality relation between light intensity and brightness. On the other hand, cosine similarity captures the angle between the vectors of values from brightness and spectrometer, without looking at the magnitude. It measures whether the variation of this angle is similar. This makes the combination of both variables complementary and useful.

3.4 Results and Discussion

3.4.1 Paper Spectral Trends

To understand the optical characteristics of each colored paper, the spectrometer measurements were obtained under white LED illumination. This allowed us to record the light intensity reflected from each paper under uniform white LED illumination, which provides a relative indication of how much light each surface reflects at different wavelengths.

Note that this is not a calibrated reflectance measurement. The data reflects a combination of the paper's reflective properties and the spectrometer's sensitivity.

This figure provides insight into which spectral regions are most reflective for each paper. These characteristics are essential to understand how much information the camera is capturing. In regions where a paper reflects minimal light, e.g. above 700 nm, the camera may struggle to record meaningful brightness values. Consequently, trend matching in those regions might result not from high fidelity but from the absence of a strong signal, leading to artificially good similarity scores.

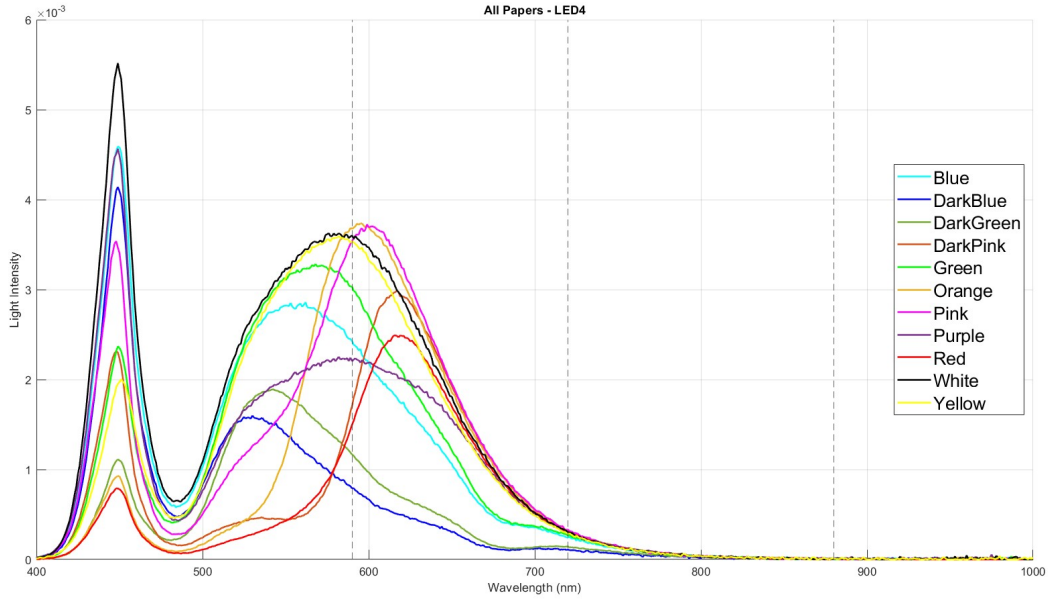


Figure 3.3: Spectral curves of all papers under white LED illumination.

3.4.2 Brightness and Spectrometer Trend Similarities

The main goal of this project was to investigate whether the brightness values recorded by a panchromatic camera under three LEDs could be used to replicate or approximate the trend of the paper's reflectance. Since the calculation of reflectance is not possible due to the lack of a calibration panel on the testbed, the trend to compare our brightness with will be light intensity as measured by the spectrometer.

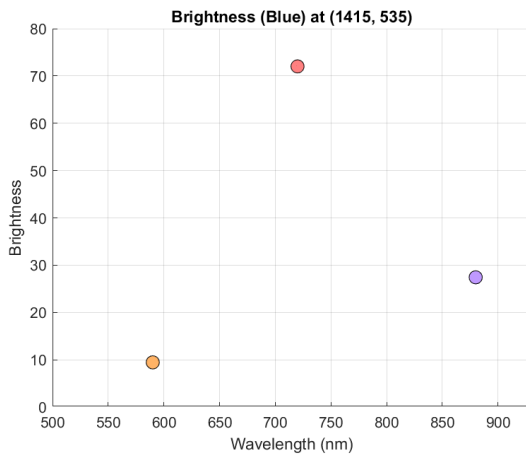


Figure 3.4: Brightness values from a chosen region of interest. Blue paper example.

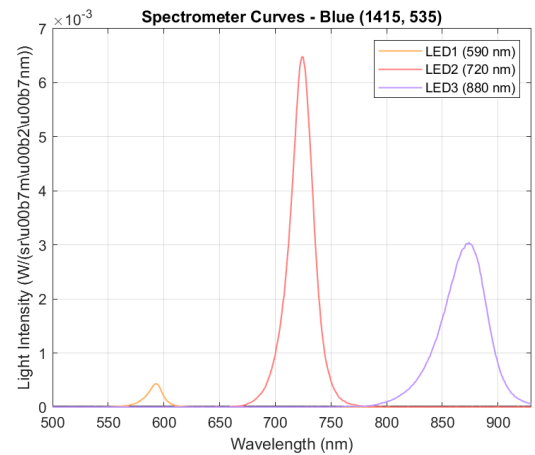


Figure 3.5: Light intensity values from a chosen region of interest. Blue paper example.

First, for each paper, the brightness values under the three LEDs were plotted against their corresponding wavelengths, as seen in Figure 3.4. These were directly compared to the spec-

trometer peak intensities at those same wavelengths as seen in Figure 3.5. This approach provides an analysis per paper, to assess whether the camera's brightness readings follow a similar trend to the actual measured spectral intensities.

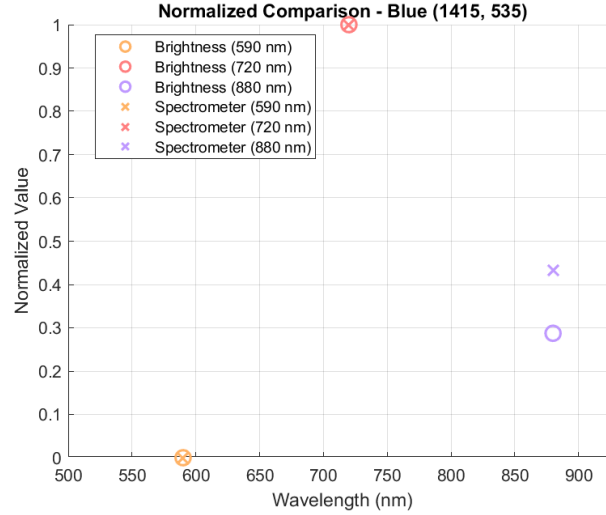


Figure 3.6: Normalized comparison between brightness values depicted on Figure 3.4 and the peak of the light intensity values depicted on Figure 3.5.

Second, an analysis per LED was performed. Here, for each LED individually, the brightness recorded for all papers was compared to their respective spectrometer values. This allows to evaluate whether a specific LED consistently reflects the same trend for all different papers, which can highlight the performance at a given wavelength. An example of these results can be observed on Figure 3.7.

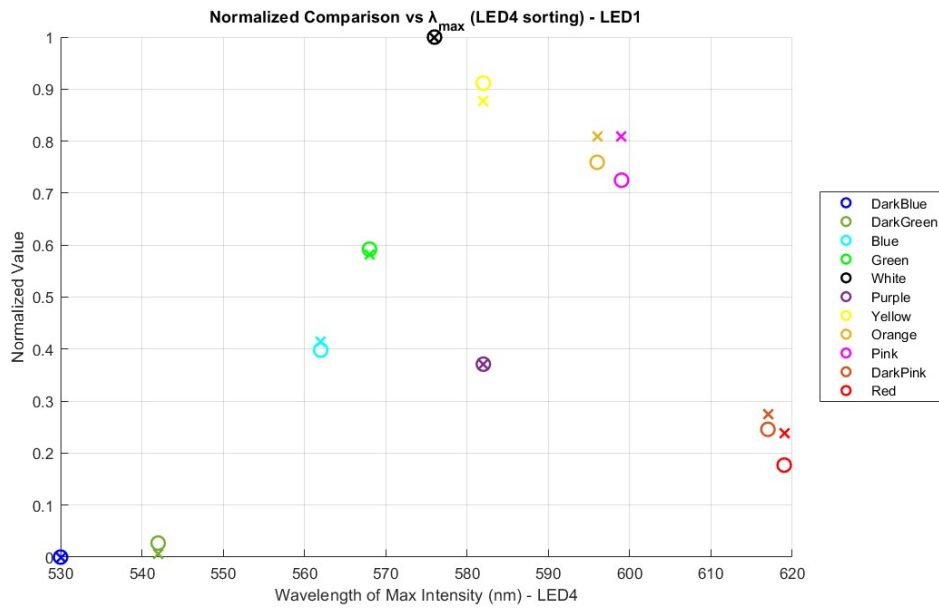


Figure 3.7: Normalized values of brightness and light intensity plotted for each paper versus their maximum on Figure 3.3 inside the panchromatic camera limit 500 to 900 nm.

Both sets of comparisons were analyzed using the Pearson correlation coefficient and cosine similarity, which are described in detail in the subsection 3.3.2. Acceptable thresholds of these correlation values depend on the case, but as a general guideline, Pearson and Cosine Similarity values above 0.95 suggest a strong similarity. Additionally, plots were computed for normalized values to more clearly observe the similarities, Figure 3.7 and Figure 3.6 as an example.

The results of the analysis per paper can be observed in Table 3.2. Most papers exhibit Pearson and cosine values above 0.98, which indicates that there is a strong correlation on the overall trends between the brightness and spectrometer values.

However, it is important to note that the best matches, such as Dark Blue and Dark Green, correspond to papers with lower intensity signal as seen in Figure 3.3. This suggests that this result may not always be due to high correlation between trends, but possibly due to the low signal to noise in those measurements. This is better observed in the trend analysis per LED.

Paper	Pearson	Cosine
Blue	0.9869	0.9953
Dark Blue	0.9979	0.9987
Dark Green	0.9972	0.9986
Dark Pink	0.9913	0.9960
Green	0.9906	0.9956
Orange	0.9879	0.99498
Pink	0.9868	0.9949
Purple	0.9919	0.9967
Red	0.9949	0.9977
White	0.9843	0.9932
Yellow	0.9864	0.9939

Table 3.2: Similarity metrics between brightness and spectrometer values for each paper.

Table 3.3 shows the similarity metrics computed across all papers for each individual LED:

LED	Pearson	Cosine
LED1 (590 nm)	0.9945	0.9980
LED2 (720 nm)	0.9800	0.9980
LED3 (880 nm)	0.9174	0.9997

Table 3.3: Similarity metrics for each LED across all papers.

It is easily observable that the yellow LED (590 nm) exhibits the strongest agreement between brightness and spectrometer data, while the infrared LED (880 nm) shows weaker Pearson correlation. This could be due to the camera's spectral sensitivity, which is lower at longer wavelengths, and reduces its ability to register brightness differences accurately in the near-infrared range. It could also be explained by the lower reflectivity of the papers in the infrared range, as shown in the spectral trends in Figure 3.3, which leads to very low intensity readings. Both of these factors can mask deviations and impact significantly the reliability of trend comparisons.

Although the Pearson correlation for the infrared LED (880 nm) is lower than that of the other LEDs, the cosine similarity remains very close to 1. This contradiction highlights the differences between the two correlation measurements. Cosine similarity measures the angle between two vectors, focusing on the general direction of change instead of the magnitude of values. As a consequence, even if both the brightness and the spectrometer measurements under LED3 are very low in magnitude and not linearly proportional, their similar trend direction can give a high cosine similarity.

The low energy recorded at this wavelength may introduce noise and reduce variability, which will affect more strongly the Pearson correlation. This suggests that while cosine similarity may overestimate the correlation of the trends under low signal conditions, Pearson offers a more conservative and scale sensitive view.

3.4.3 Paper Recognition

Once this method was implemented and its feasibility studied, it was necessary to evaluate whether a reverse approach was viable. Identifying the color of an unknown paper based only on its brightness measurements, by comparing them to the available spectrometer data from all known samples.

Several scripts were developed to compute the similarity between the brightness of a chosen paper and the spectrometer intensity trends for each reference paper. However, this backwards approach often failed to identify the paper.

The first approach consisted on identifying the paper by computing its Pearson and Cosine correlation with all the spectrometer values. However, this lead to inaccurate results since the brightness and spectrometer correlation with the incorrect paper could be higher, due to the trend similarities.

In the second approach, the identification of the correct paper is based on how its brightness and spectral intensity differ from a white reference. The idea is that each paper reflects and absorbs light differently compared to a white surface, and this difference can be used as a unique signature. Our reference in this case was the white paper. This method is not ideal as it is not applicable on Phobos. This method gave slightly better results, however the identification of the paper remained mostly inaccurate.

The third approach consisted on computing the ratios between brightness values and comparing them to the corresponding ratios from the spectrometer data, e.g. computing the ratio of the brightness at 590 nm and at 720 nm and comparing the same ratio with the spectrometer. By comparing these relative ratios, the method becomes independent of absolute intensity or brightness levels and focuses instead on the pattern of reflectance behavior across wavelengths. Once again, for this method, the results were not conclusive.

This could be a result of most colored papers exhibiting a highly similar spectral behavior as observed in Figure 3.8, lowest reflectance under LED1 (590 nm), intermediate under LED3 (880 nm), and highest under LED2 (720 nm). This pattern makes it difficult to distinguish the papers based solely on the direction or shape of their response curves since both cosine similarity and Pearson correlation measure directional or trend similarity only, but do not

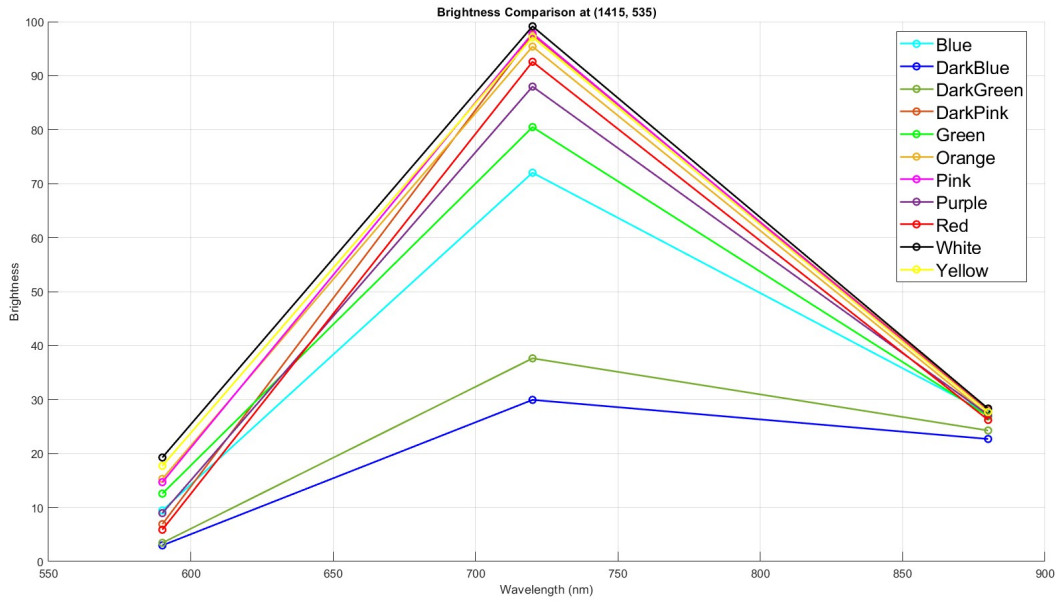


Figure 3.8: Comparison of brightness trends per paper

measure closeness numerically. Magnitude is completely disregarded in these approaches, which can lead to incorrect results, since it is possible for an incorrect paper to have a similarly shaped response vector but very different absolute values.

In addition, there could be small variations in the image brightness due to ambient light, which would cause the panchromatic camera to receive more signals other than the LED, which could have contaminated the results.

Interestingly, when analyzing the spectral measurements of the papers under white LED illumination in subsection 3.4.1, it was expected that the papers showed a higher reflectance in the yellow region (closer to LED1). However, the data collected indicated a stronger reflectance under LED2. This discrepancy could most likely be due to differences in the emitted power of the LEDs. If LED2 emits light with more power than LED1, it could cause all papers to appear brighter under LED2 illumination, even if their reflectance is naturally higher in the yellow range. This makes comparisons with known spectral curves more difficult, particularly if those curves are not derived under the same lighting conditions. Additionally, this approach relies on relative measurements that are susceptible to environmental conditions and assumptions regarding uniform illumination and camera response. Next semester, a calibration panel will be added to our measurements in order to compute values of reflectance. This will add homogeneity to our measurements and, ideally, improve our results.

Furthermore, the comparison relies solely on three measurements per paper, one for each LED. This set of data is of a very low dimension and therefore, it is highly sensitive to noise and lacks enough detail to reliably differentiate between materials with similar trends but different reflectance magnitudes. This method does not incorporate the actual scale of the measurements, it only looks at the relative shape, which means it struggles to differentiate between papers unless their trends are noticeably distinguishable. This highlights the limitations of this method

and the importance of assessing the magnitude for future tests.

3.4.4 Assumptions

Here, a complete summary of the assumptions necessary to simplify the experimental approach. First, each LED was assumed to emit light at a single narrow wavelength, despite its known slight variation. This allows for a simplified analysis, though it introduces inaccuracies in the spectrometer measurements, where an offset at the peak position was observed at each wavelength.

Additionally, the sensitivity factor of the camera was assumed to be constant for each LED wavelength and assumed to have been corrected in the cameras calibration through the spectral response values provided by the manufacturer. This assumption will be verified once the calibration of our measurements is made.

Moreover, it was assumed that the selected point, identified by the spectrometer's laser receives approximately equal and sufficient illumination from all LEDs. The angular response of the camera was not explicitly considered, based on the assumption that it would not significantly affect the behavior of the results under the simplified setup.

Note that the difference in optics as explained in subsection 1.1.4 between the WheelCams on IDEFIX and the testbed WheelCams will be substantial, and will have to be considered when the complexity of the model evolves to a proxy of our Phobos environment.

Lastly, the main assumption is that there is a linear relationship between image brightness and material reflectance. This is reasonable for our simplified model, but it may not hold in more complex or uncontrolled conditions.

3.5 Future Work

The next steps of this research will focus on improving the experimental model to transition from the simplified paper samples to more representative geological materials, such as basalt, olivine, and pyroxene. These samples will allow a more realistic assessment of the differentiation capabilities of the combination between panchromatic cameras and colored LEDs. In Figure 3.9, one of the proposed basalt samples can be observed, with visible olivine crystals and black pyroxene crystals.

Revisiting the recognition step of our experiments will be a priority. Either refining the scripts that already exist to improve the algorithms or writing a new one for the following experiments with materials. Furthermore, the correlation coefficients used will be reevaluated and changed if deemed necessary.



Figure 3.9: Mantle enclaves (ol+cpx+opx), isolated olivine (green) and pyroxene (black) large crystals in lavas from Beaunit (Auvergne)

A key step will be the integration of a calibrated reflectance standard into the image acquisition process. This addition will enable the conversion of brightness values into absolute reflectance measurements, allowing for more accurate trend comparisons. A calibrating panel will be used as reference during every experimental measurement, and its values will be incorporated into the post processing for normalization. Additionally, this approach will help improve the readings affected by the intensities of the LEDs. Since reflectance will be calculated, this effect will no longer be visible on our results.

Further development will involve the including the characterization of the LED emission spectra. This was done in the past and by including it in future characterization measurements such as the ones performed in subsection 3.4.1, normalized values of the samples' behavior under each LED can be obtained. This will improve the correlation between camera and spectrometer measurements.

4 Conclusion

In this research, we contributed to the scientific preparation for the MMX mission by developing image processing tools that support the analysis of WheelCam images from the IDEFIX rover. Specifically, we focused on advancing two core objectives: estimating wheel sinkage and exploring the potential for spectral analysis using panchromatic images illuminated by narrow-band LEDs. To achieve this, we improved existing algorithms for wheel detection and ellipse fitting, enabling more accurate identification of the wheel-regolith interface across varying conditions. Additionally, we developed a simplified framework to assess whether spectral information about the regolith could be inferred from images taken under different illumination wavelengths. These tasks were facilitated by using a dedicated testbed designed to simulate the conditions on Phobos, allowing us to validate our methods in a controlled environment. Through this work, we laid essential groundwork for supporting key scientific goals of the MMX mission, particularly in understanding the mechanical and compositional properties of Phobos' regolith.

Wheel sinkage

The key conclusions regarding the sinkage estimation part of this research can be summarized as follows. The algorithm for detection the wheel contour and fitting an ellipse to it has demonstrated strong performance when applied to images from the rear WheelCam under varying test conditions. Notably, its accuracy improves when used with the JAXA regolith simulant, which is expected to resemble the surface conditions on Phobos most closely. However, performance of this algorithm on the front WheelCam images remains suboptimal, indicating a need for further improvements in that area.

The next major steps needed to progress this research are presented in the following task breakdown:

1. Validation of wheel detection algorithm for rear WheelCam images. This includes i) testing for different SNR levels and ii) testing on a wider range of regolith types and sinkage levels
2. Completion of sinkage estimation pipeline. This includes i) applying a homography transform to the detected ellipse ii) detecting the wheel/regolith interface and iii) calculation of sinkage level.
3. Definition of performance requirements for the complete wheel sinkage algorithm. This includes i) defining the required sinkage estimation accuracy and ii) derive subsequent requirements on contour quality and ellipse accuracy
4. Optimization of wheel detection algorithm for front WheelCam images. This includes i) tweaking model parameters of the existing algorithm or investigating other segmentation strategies and ii) validating this algorithm

Spectral studies

This study has demonstrated the feasibility of detecting material composition using a simplified experimental model based on panchromatic image brightness under LED illumination. A clear correlation was found between brightness and light intensity values, where for most colored papers, high Pearson correlation and cosine similarity were obtained, as seen on Table 3.2. This indicates that the brightness trends reliably reflected the spectral behavior of the colored papers.

However, several limitations were identified. The samples' reduced reflectance at longer wavelengths, especially under the 880 nm infrared LED, affected the possibility of accurately measure brightness values. For the future, a sample with visible reflectance or absorption peaks on the near infrared should be considered.

Because the samples consisted of smooth colored paper, only a single region of interest was analyzed for each one. While this is reasonable for homogeneous surfaces, it would be insufficient for natural or irregular materials with more complex texture or grain variation.

Additionally, the absence of a calibrated reflectance measurement limited the possibility of absolute material identification. The attempts to identify materials based only on brightness trends proved to be inconclusive, largely due to the similarity between spectral trends of the different colored papers.

Finally, the number of data points per sample is inherently limited to three (one per LED), which constrains the information obtained and hinders the ability to reliably identify correlations between trends. This is not a limitation of the simplified model per se, but rather of the hardware constraints of the IDEFIX rover system, and therefore must be accepted as a boundary condition for future development.

Despite these challenges, the experiment successfully validated our initial hypothesis: under controlled lighting and simplified conditions, panchromatic image brightness represents meaningful information related to material reflectance. This provides a foundation for more advanced material composition detection methods using the imaging systems aboard the IDEFIX rover.

The following tasks outline the next steps detailed in section 3.5 to advance the material composition detection portion of this research:

1. Transition to geologically representative materials. This includes: i) replacing the colored paper samples with materials such as basalt, olivine, and pyroxene, and ii) acquiring brightness and spectrometer data from these new samples under the existing LED setup.
2. Improvement of material recognition algorithms. This includes: i) refining or redesigning the material recognition algorithm to improve matching accuracy and ii) evaluating alternative similarity metrics to replace or complement Pearson and cosine coefficients.
3. Integration of calibrated reflectance reference. This includes: i) incorporating a standard calibration panel into all future image acquisitions, and ii) using this panel to normalize brightness values and convert them into reflectance estimates during post-processing.
4. Apply LED spectral characterization data to correct for variations in LED output and improve consistency between camera and spectrometer measurements.

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